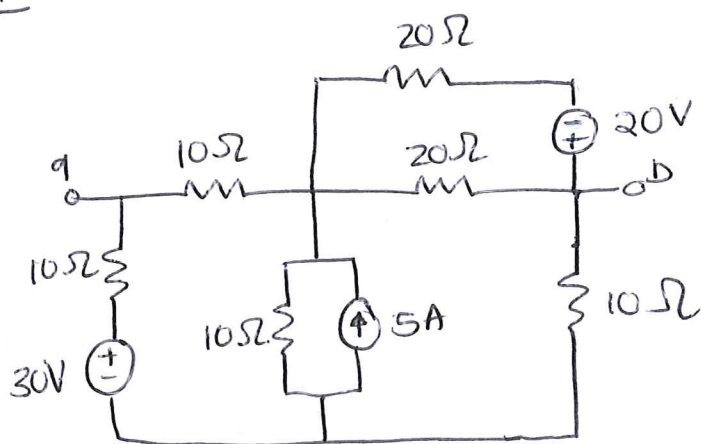


Tutorial 5

①

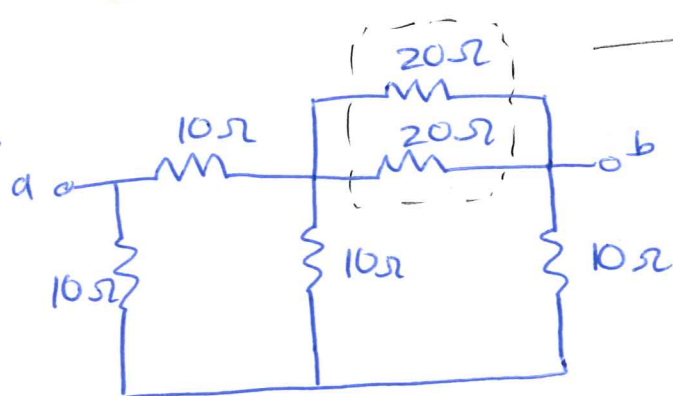
4.42



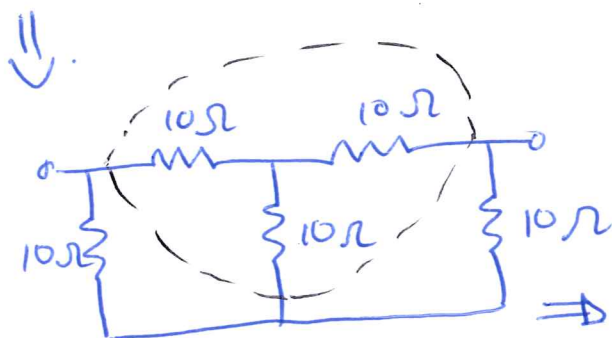
$V_{TH} ?$

$R_{TH} ?$

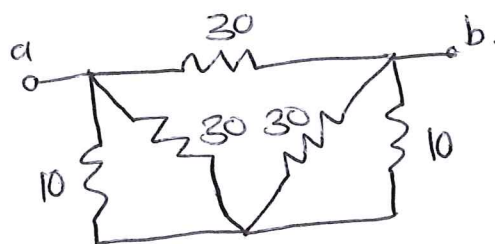
① R_{TH} → switch off all INDEPENDENT sources



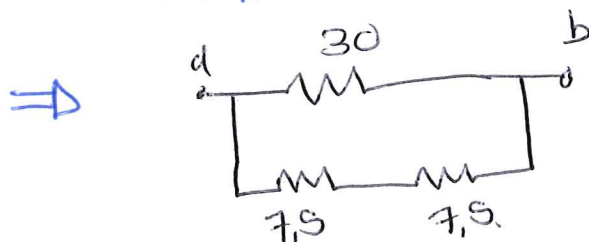
— $20 \parallel 20 = 10 \Omega$



$Y \Rightarrow \Delta \therefore 3Y = \Delta$
 $R = 30 \Omega$



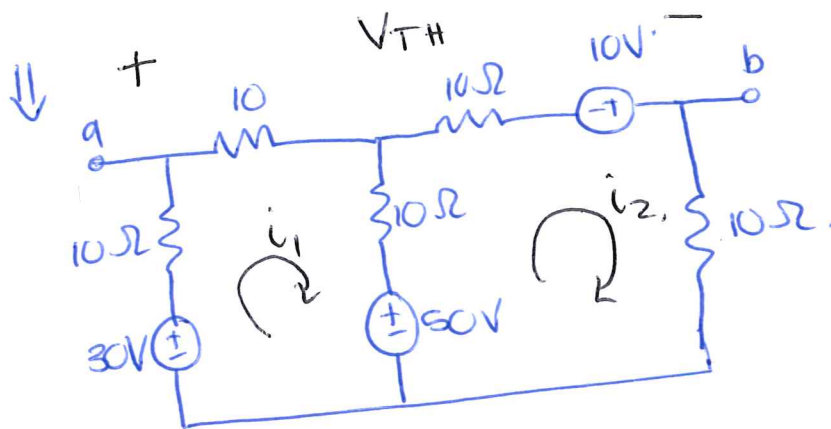
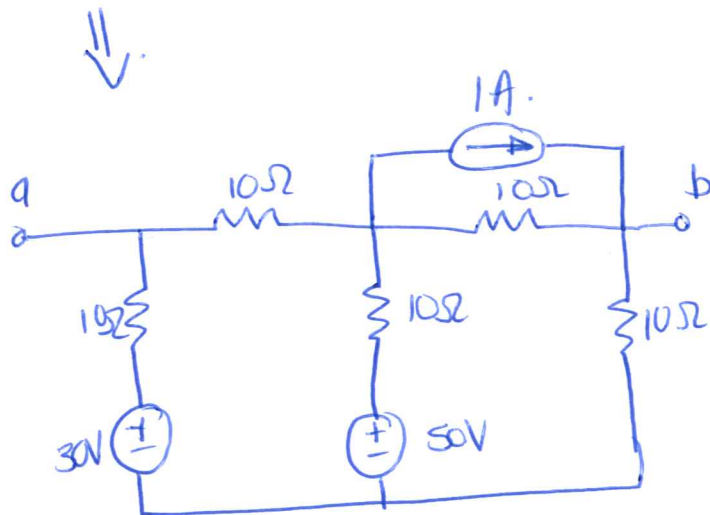
$\times 10 \parallel 30 = 7,5 \Omega$



$\therefore R_{eq} = R_{TH} = 30 \parallel 15 = 10 \Omega$

②. V_{TH} → Use SOURCE TRANSFORMATION

②



$$\begin{aligned} \underline{L1}: -30 + 30i_1 - 10i_2 + 50 &= 0 \\ 30i_1 - 10i_2 &= -20 \\ 3i_1 - i_2 &= -2 \end{aligned} \quad \text{--- (1)}$$

$$\begin{aligned} \underline{L2}: -50 + 30i_2 - 10i_1 - 10 &= 0 \\ -i_1 + 3i_2 &= 6 \\ \Rightarrow i_1 &= 3i_2 - 6 \end{aligned} \quad \text{--- (2)}$$

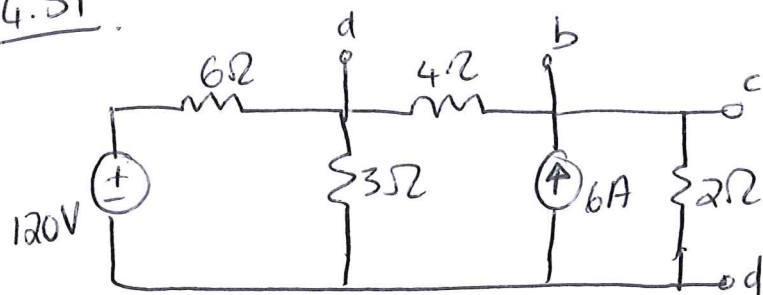
② in ①

$$\begin{aligned} 3(3i_2 - 6) - i_2 &= -2 \\ 9i_2 - 18 - i_2 &= -2 \\ 8i_2 &= 16 \\ \underline{i_2} &= 2 \\ \therefore \underline{i_1} &= 3(2) - 6 = 0 \end{aligned}$$

⇒ $V_{TH} = V_{ab}$

$$\begin{aligned} V_{TH} &= 10(i_1) + 10(i_2) - 10 \\ &= 20 - 10 \\ &= 10V \end{aligned}$$

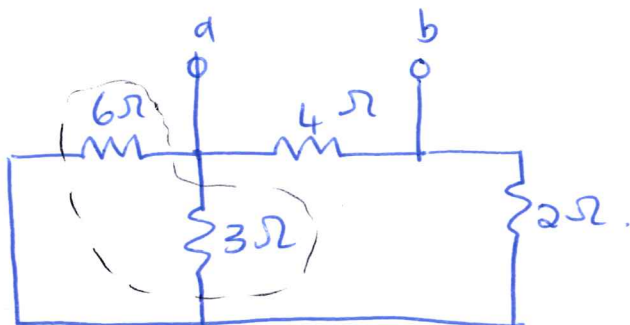
4.51.



Norton equivalent
as viewed from:

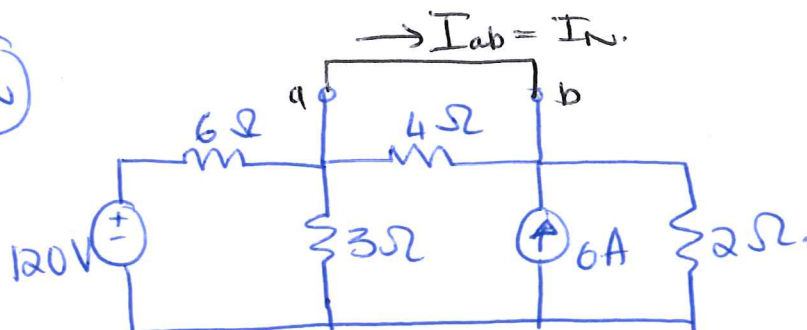
- a) a-b
- b) c-d.

a
 R_N



$$\begin{aligned}
 R_N &= [(6 \parallel 3) + 2] \parallel 4 \\
 &= (2 + 2) \parallel 4 \\
 &= 2\Omega
 \end{aligned}$$

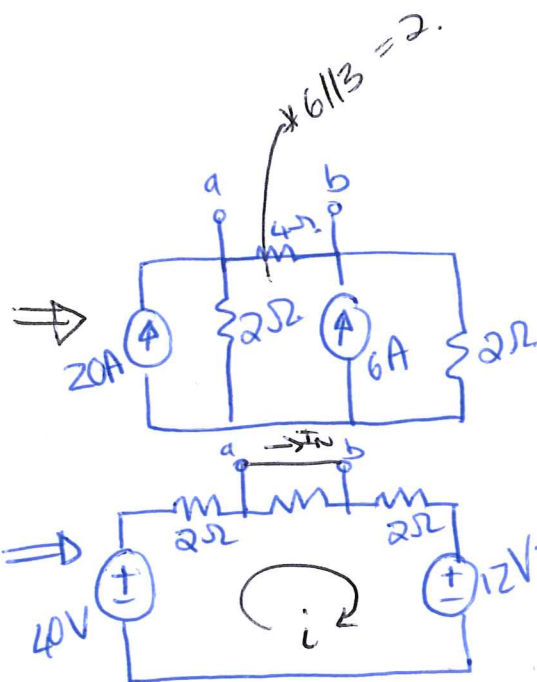
I_N



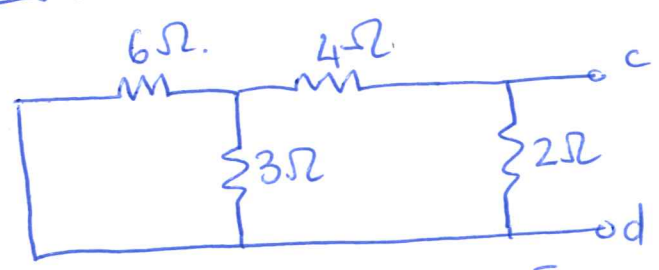
$$i = I_N$$

$$\begin{aligned}
 \therefore -40 + 4i + 12 &= 0 \\
 i &= \frac{28}{4} = 7
 \end{aligned}$$

$$\therefore \underline{I_N = 7A}$$



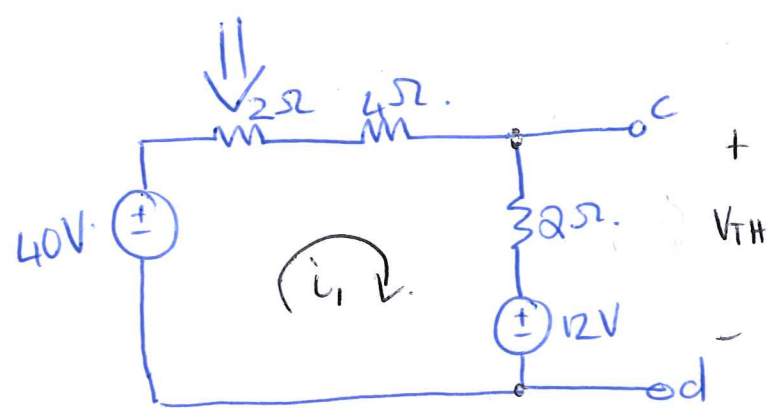
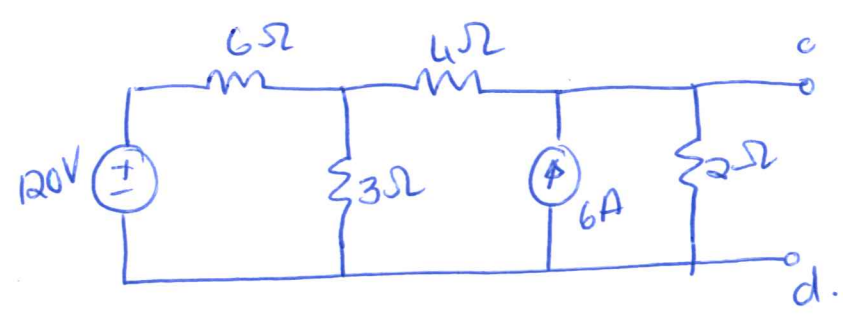
b) R_N



$$R_{eq} = R_N = \left[(6/3) + 4 \right] // 2$$

$$= 1,5 \Omega$$

I_N



$$\rightarrow -40 + 6(i_1) + 2(i_1) + 12 = 0$$

$$8i_1 = 28$$

$$i_1 = 3,5 \text{ A}$$

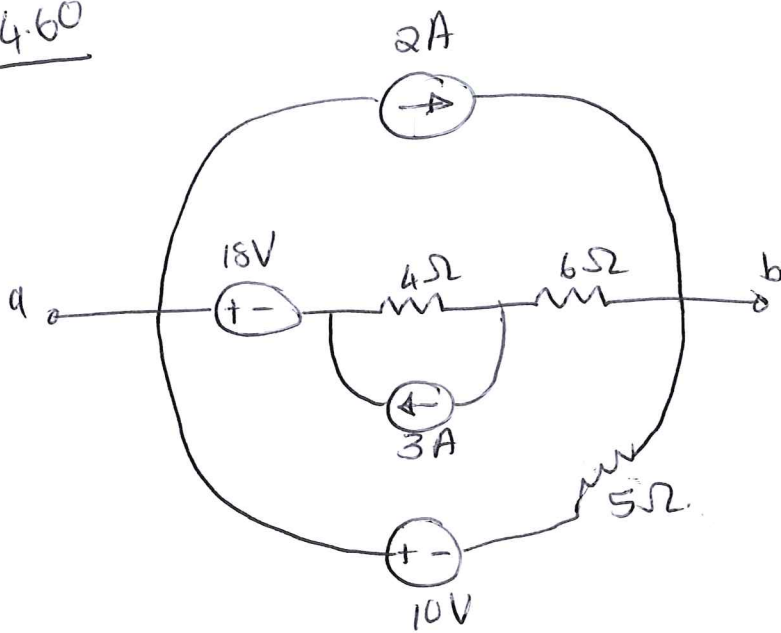
* V_{TH} is the voltage across 2Ω -resistor and $12V$ -source.

$$\therefore V_{TH} - 12 = 2(3,5) \rightarrow V_{TH} = 19 \text{ V}$$

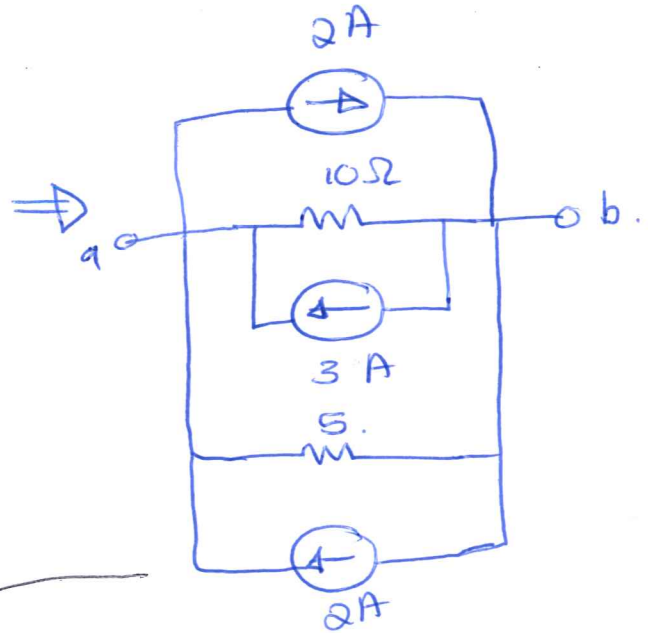
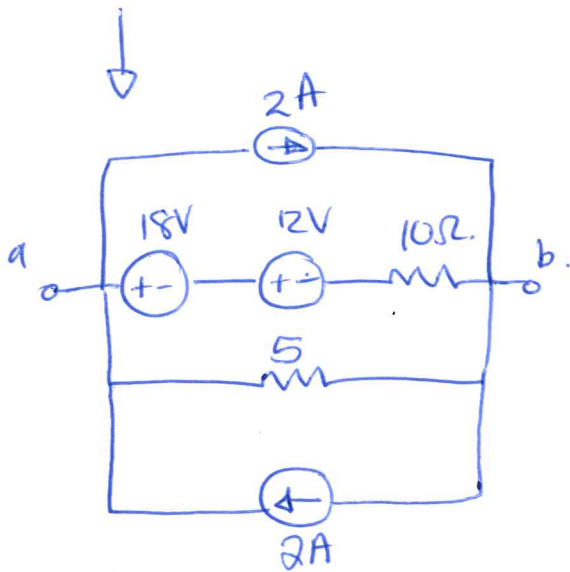
$$\therefore I_N = \frac{19}{1,5} = 12,67$$

4.60

8

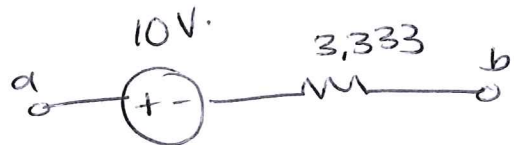
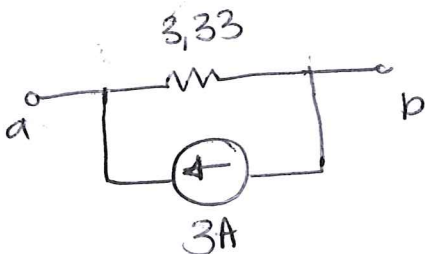


Thevenin and Norton equivalent at a-b.

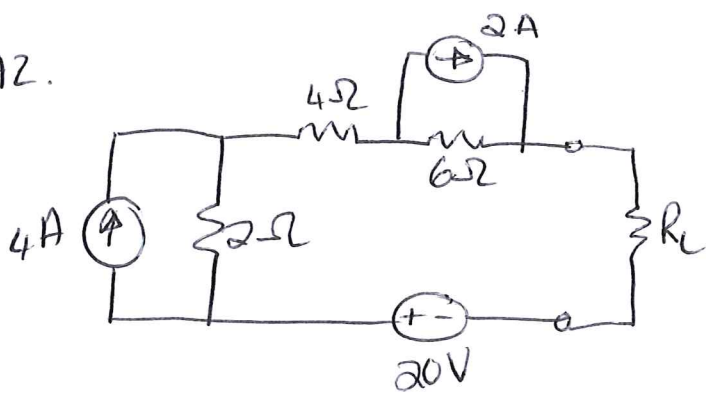


Norton Circuit.

Thevenin Circuit



4.72.



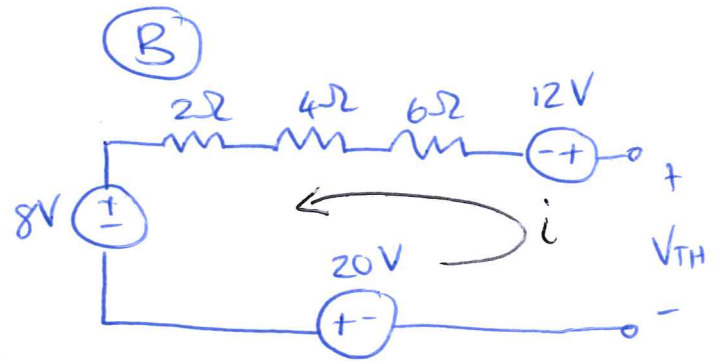
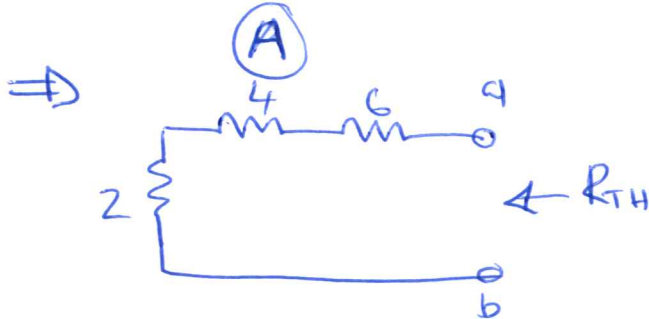
a) V_{TH} , R_{TH}

b) i_{RL} ? $R_L = 13\Omega$

c) R_L for max power

d) P_{max}

(12)



a) • $R_{TH} = 2 + 4 + 6 = \underline{12\Omega}$, (Circuit A)

• V_{TH} (Circuit B)

$$-V_{TH} + 12 + 8 + 20 = 0$$

$$\underline{V_{TH} = 40V}$$

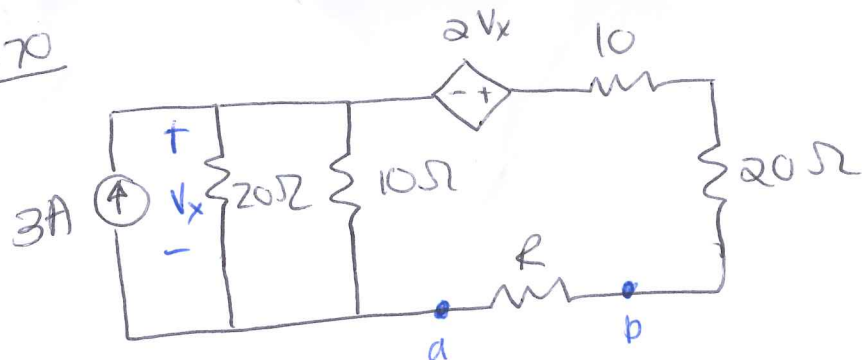
* V_{TH} is OS-voltage, thus $i = 0$.

b) $i_{RL} = \frac{V_{TH}}{R_{TH} + R_L}$
 $= \frac{40}{12 + 13}$
 $= \underline{1.6\text{ A}}$

c) For max power transfer: $R_L = R_{TH} = \underline{12\Omega}$

d) $P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{40^2}{4(12)} = \underline{33.33\text{ W}}$

4.70

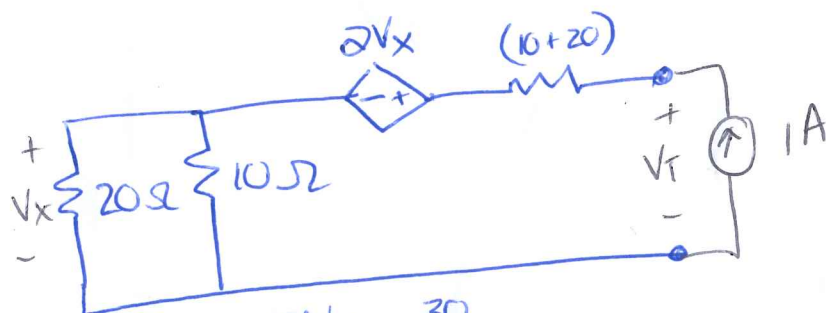


$P_{\max}$ delivered to R .

(10)

→ Determine Thevenin equivalent

R_{TH}



$$\rightarrow -V_t + 30(1) + 2V_x + V_x = 0$$

$$V_t = 30 + 3V_x$$

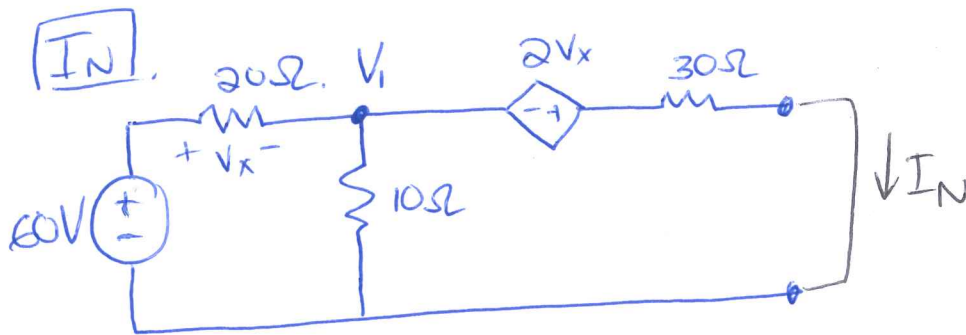
$$\times V_x = 6.6 - (1) = 6.1$$

② in ①: $V_t = 30 + 3(6.6)$
 $= 50.00001$

$$\therefore R_{TH} = \frac{V_t}{I_t}$$

$$= 50 \Omega$$

(11)



$$* V_x = 60 - V_1$$

$$\rightarrow \frac{60 - V_1}{20} = \frac{V_1 - (-2V_x)}{30} + \frac{V_1}{10}$$

$$180 - 3V_1 = 2V_1 + 4(60 - V_1) + 6V_1$$

$$180 - 3V_1 = 8V_1 + 240 - 4V_1$$

$$-7V_1 = 60$$

$$V_1 = -8,57 \text{ V}$$

$$\rightarrow I_N = \frac{V_1 - (-2V_x)}{30}$$

$$I_N = -4,28 \text{ A}$$

$$\Rightarrow \therefore V_{TH} = I_N R_{TH} \\ = (-4,28)(50) \\ = -214 \text{ V}$$

$$\therefore P_{max} = \frac{V_{TH}^2}{4R_{TH}} \\ = \frac{(-214)^2}{4(50)} \\ = 228,98 \text{ W}$$

Tutorial 5

$$4.42) R_{TH} = 10\Omega ; V_{TH} = 10V$$

$$4.47) R_{TH} = R_N = 20\Omega ; V_{TH} = 20V ; I_N = 1A$$

$$4.51) \text{ [a] } R_N = 2\Omega ; I_N = 7A$$

$$\text{ [b] } R_N = 1,5\Omega ; I_N = 12,67A$$

$$4.53) R_N = 10\Omega ; I_N = +3A$$

$$4.60) R_N = R_{TH} = 3,333\Omega ; I_N = 3A ; V_{TH} = 10V$$

$$4.64) R_{TH} = -1\Omega ; V_{TH} = 0V$$

$$4.72) \text{ [a] } R_{TH} = 12\Omega ; V_{TH} = 40V$$

$$\text{ [b] } i_{RL} = 1,6A$$

$$\text{ [c] } R_L = R_{TH} = 12\Omega$$

$$\text{ [d] } P_{max} = 33,33W$$

$$4.73) R_{TH} = 10,833\Omega ; V_{TH} = 30V ; P_{max} = 20,77W$$

$$4.75) R_L = 250\Omega$$

$$4.70) R_{TH} = 50\Omega ; V_{TH} = -214V ; P_{max} = 228,98W$$